

Estimating pollen diversity from pollen accumulation rates: a method to assess taxonomic richness in the landscape

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Abstract: A new method is presented to estimate pollen diversity on the basis of pollen-accumulation rates, rather than on a constant count of pollen as is traditional. Accumulation-based estimates of pollen diversity can be interpreted in terms of taxonomic richness on the landscape. The traditionally used constant-count estimates, on the other hand, reflect the evenness in the vegetation much more than its taxonomic richness, but they reflect this in a distorted way because of the real but mostly unknown differences in pollen productivity among plant taxa. Pollen-accumulation rates might also be used to estimate turnover of plant species from pollen assemblages. Differences in pollen productivity among plant taxa also have effects on various other data-summarizing techniques, including estimates of rates of change.

Key words: Pollen diversity, taxonomic richness, pollen accumulation rates, pollen count, pollen productivity, plant species turnover.

Introduction

Since Birks and Line (1992), rarefaction analysis of pollen assemblages has frequently been used in palaeoecology to estimate palynological richness, which is then interpreted in terms of plant diversity (species richness) in the landscape. Rarefaction analysis estimates the number of pollen types that would occur in each pollen sample if the same number of pollen grains had been counted per sample: the rarefaction sum. (This must be equal to or smaller than the sum actually counted.) Birks and Line (1992) discussed in detail the background assumptions, properties and pitfalls of the method and provided examples from five pollen sequences with plausible interpretations. Yet one potential pitfall has worried me ever since, namely the effect of evenness of the pollen assemblage on the estimated diversity, and I am not alone in these worries (eg, Odgaard, 1999, 2001: 'Palynological diversity is undoubtedly a reflection of vegetation diversity but the relationship between the two is still incompletely understood').

The palynological diversity as estimated by rarefaction analysis seems to be a mixed expression of two components: evenness in the pollen assemblage and plant diversity in the landscape. The relative strength of the two signals has remained elusive. This needs further exploration and methods need further development, as it is species richness in the field that most palynologists wish to estimate from pollen, independent of pollen evenness. Odgaard (2001) found a positive correlation between pollen richness and pollen evenness. Peros and Gajewski (2008) found the same in the Canadian Arctic and Greenland and, even more disconcerting,

they found a negative correlation between pollen richness and species diversity in the landscape, which they ascribed to the equally negative correlation between pollen evenness and species diversity in their field situation.

These and other studies show that the traditionally determined pollen diversity simply does not work when we aim at taxonomic diversity in the landscape, and to all appearances it is the differences in evenness among pollen assemblages that distort the signal. To solve this problem we need a technique of estimating pollen richness that is independent of pollen evenness. Here I present a new method of rarefaction analysis that is based on pollen accumulation and thus excludes the effect of evenness completely. This method answers the question: how many different pollen types are deposited every year on a defined, constant surface? The aim of this paper is to provide this essential but, to date, missing methodological step in estimating taxonomic richness in the landscape from pollen assemblages.

Methods and results

The pollen data set is derived from Gerzensee on the Swiss Plateau; see Ammann (2000) for a description, and B. Ammann, W. van der Knaap and J. van Leeuwen (in preparation, 2008) for a full report of the research of which this note is an outcome.

Results are shown in Figure 1. For pollen-diversity estimates, all pollen types in the pollen sum were used. Pollen diversity was estimated in two ways: (A) on a *constant count* of pollen, being the number of pollen types that would have been encountered if all pollen samples were counted to the same pollen sum; and (B) on *pollen accumulation*, being the number of pollen types deposited each year on a defined sediment surface.

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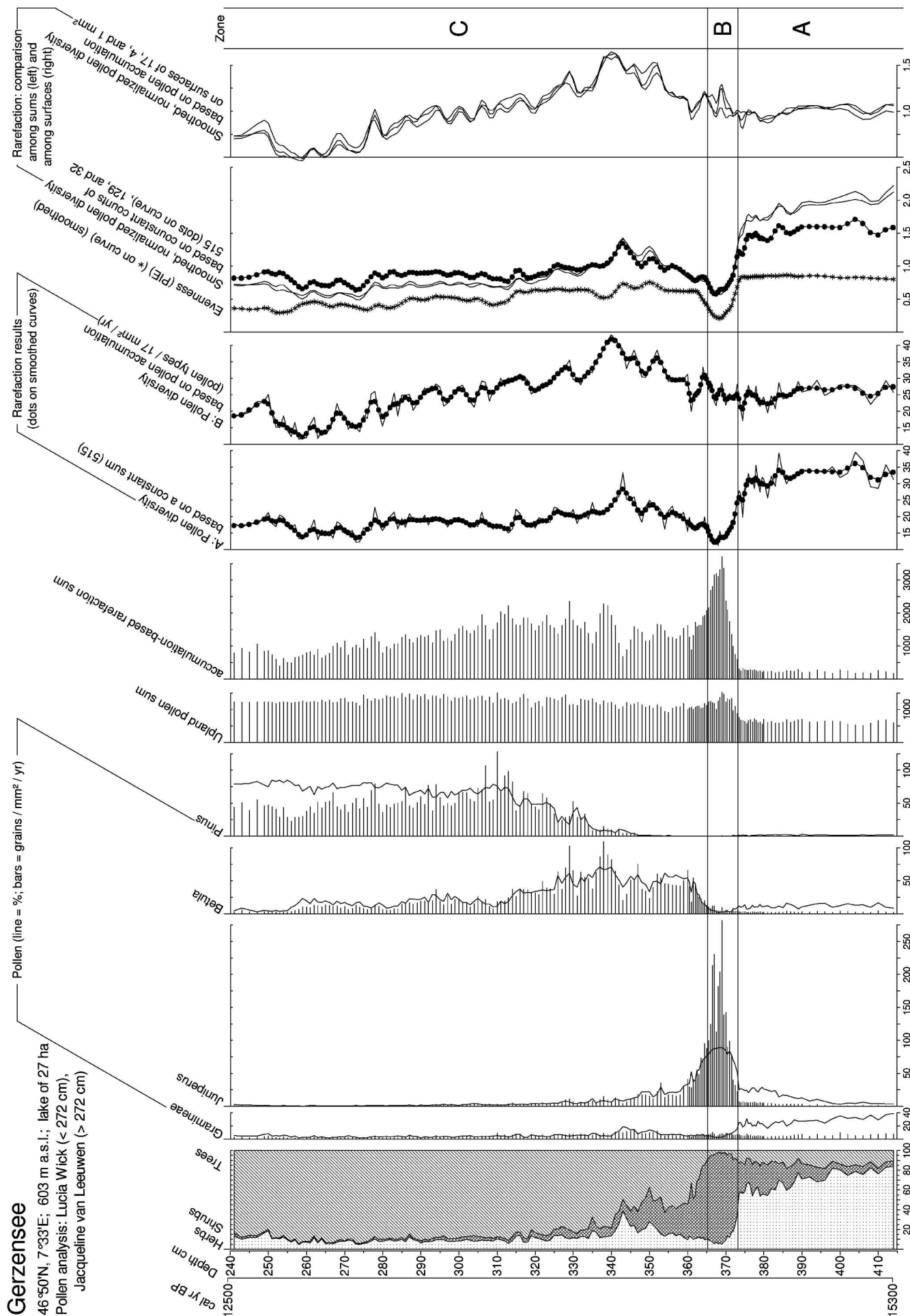


Figure 1 The example data set: selected pollen curves and results

(A) Estimating pollen diversity on a constant count was carried out with the rarefaction software RAREPOLL (Birks and Line, 1992). Results are smoothed with a weighted running mean over five samples (Figure 1).

(B) Estimating pollen diversity on pollen accumulation was more complicated, as no user-friendly software is available. The additional calculations can be done using TILIA software (Grimm, 1990) and Excel. It involved the following steps:

- Calculate for each pollen sample the *upland pollen sum* (shown in Figure 1) (in TILIA).
- Calculate for each pollen sample the *total pollen accumulation* (grains/cm² per yr); this requires estimation of pollen concentration using an exotic marker and a depth–age model to convert pollen concentrations into pollen-accumulation rates (in TILIA).
- Calculate for each pollen sample the *surface* on which the *upland pollen sum* is accumulated annually (cm²; not shown), by dividing the *upland pollen sum* by the *total pollen accumulation* (in Excel).
- Search for the *smallest surface* among the *surfaces* determined above. Some pollen samples in our data set had tiny surfaces, in which case three consecutive pollen samples were added together (in running sums) in order to reach a larger *smallest surface* in the data set. In our data set, this resulted in a *smallest surface* of 0.17 cm².
- Calculate for each pollen sample the *number of pollen grains* deposited annually on the *smallest surface* (grains/smallest-surface per yr), by multiplying the *smallest surface* with the *total pollen accumulation* (in Excel) (in Figure 1 shown as grains/17 mm² per yr). This resulted in nearly as many different answers as there are pollen samples in the data set (some samples resulting in the same answer).
- Run the rarefaction software RAREPOLL for each pollen sample separately on the basis of the *number of pollen grains* previously obtained for that sample.
- Assemble the resulting *diversity estimates* in a stratigraphic sequence. Results were smoothed with a weighted running mean over five samples (Figure 1).

In estimating pollen diversity on the basis of a constant count, the main reason to choose the highest possible count (here, 515) is not so much for obtaining the highest possible pollen-diversity estimates, but rather because we know from experience that the stratigraphic trend in pollen diversity may shift when estimated on a very low pollen sum. To show this, two alternative pollen-diversity estimates were added, one for a reduced pollen sum of 25% of the maximum size (129) and another for a further reduced sum of 25% of the latter (32). Each curve was smoothed with a weighted running mean over five samples and normalized by dividing the diversity estimates by their long-term average, and the three curves were collected in a single graph, so that the trends in the curves can be compared directly (Figure 1). Similarly, the effect of the size of the accumulation surface on accumulation-based diversity estimates is explored by using three different surfaces in equally large steps of $4 \times$ (17, 4 and 1 mm²), followed by smoothing and normalizing the curves and printing them together as in the other diversity estimate (Figure 1). The average annual pollen accumulation on 1 mm² is 74 (range 10–319).

Evenness of the pollen samples was calculated according to Hurlbert's (1971) *probability of interspecific encounter* (PIE): the probability that two randomly selected pollen grains (without replacement) will be of different taxa. Peros and Gajewski (2008) also chose this measure of evenness, as it is neither affected by sample size nor by pollen diversity, unlike several other evenness

measures. The theoretical limits of PIE are 0 when all pollen grains in a sample are the same and 1 when they are all different. The evenness is shown as a smoothed curve (weighted running mean over five samples) in the graph that also shows the three diversity curves based on different constant sums.

Potential pitfalls in accumulation-based estimates of pollen diversity

The accumulation-based pollen-diversity estimates have a few potential sources of error (see discussion in Birks and Line, 1992). First, they have all the errors that the estimates of pollen-accumulation rates (grains/cm² per yr) have, such as those caused by sediment focusing, errors in the depth–age model, and errors in the laboratory procedures (sample-volume measurement, statistical errors in both the added and the counted exotic marker). Second, differences among pollen analysts in their ability to distinguish the less common pollen types could have a major impact on the estimates, owing to the absence of the effect of evenness.

Interpretation of pollen-diversity estimates

The two pollen-diversity estimates show both differences and similarities. The main difference is in Zone B, where the phase of abundant *Juniperus* pollen marks the transition from shrub tundra to open woodland. The diversity estimate based on pollen accumulations suggests that taxonomic diversity in the landscape was about similar in Zones A and B, whereas the diversity estimate based on a constant count shows a marked drop from Zone A to B that follows the drop in evenness of the pollen assemblage caused by the *Juniperus* expansion. Based on a constant count, the estimated diversity is distinctly lower in Zone C than in Zone A, but based on pollen accumulation the two zones have about similar values. Nevertheless, within these zones the two estimates of diversity have similar general trends. Thus the two estimates of pollen diversity apparently tell different stories.

The taxonomic plant diversity in the surrounding landscape is just one of the various factors determining pollen diversity (see Weng *et al.*, 2006 for a discussion). When based on a constant count, pollen diversity also depends on the evenness of the pollen assemblage; in our example the two correlate with $R = 0.844$ (Pearson correlation of unsmoothed values). Low evenness (ie, large differences in abundance among pollen types) lowers a pollen-diversity estimate based on a constant count (examples in, eg, Räsänen *et al.*, 2004), because the high influx of abundant pollen types diminishes the chance of finding rare pollen types. This is exemplified by the *Juniperus* maximum in Zone B and the subsequent *Betula* and *Pinus* dominance in Zone C. These three taxa are large pollen producers compared with other plants in the Lateglacial landscape, and when they start flooding the sediment with their pollen the chance of encountering many other pollen types is diminished if counting stops at a defined pollen sum.

Pollen diversity based on a constant count depends on both evenness and plant diversity in the landscape. When large changes in dominant pollen types occur in the pollen diagram, evenness seems to dominate the trend in pollen diversity based on a constant count and plant diversity seems subordinate. We can explain the main features in the constant-count pollen-diversity curve by comparing the curve with the curves of dominant

pollen types. The positive correlation of constant-count pollen-diversity with total herb pollen is striking ($R = 0.908$ in Pearson correlation with unsmoothed values). However, the constant-count pollen-diversity curve alone does not give a clue which of the two, evenness and plant diversity, determines which features. Summarizing, we can *explain* some features in the constant-count pollen-diversity curve in terms of dominant pollen types, but we cannot *interpret* the curve in terms of plant diversity in the vegetation – note the difference between explaining and interpreting; only the latter generates new information. Floristic evenness in vegetation is of interest in itself, so it would be worth estimating it from pollen. However, differences in pollen productivity among taxa modify the evenness in the pollen assemblages compared with the evenness in the vegetation, and this modifies the reflection of evenness in any measure, including constant-count pollen diversity. Constant-count pollen diversity, though interesting as a method, is therefore inadequate for estimating or interpreting plant diversity.

Our test of robustness of the two measures of pollen diversity, using three different pollen sums for the constant-count pollen diversity and three different surfaces for the accumulation-based pollen diversity, emerges in favour of the latter (Figure 1, the two right-hand graphs). The impact of evenness on constant-count pollen diversity appears to increase with decreasing pollen sums, whereas the decrease in accumulation surfaces does not show any impact on accumulation-based pollen diversity. This allows simplifying the method of estimating accumulation-based pollen diversity, because the accumulation surface can be standardized to a low value that is reached in all sites of a specific study. A worldwide standard of, eg, 1 mm² can however not be accepted, because in some geographic regions with generally low pollen-accumulation rates (such as the high arctic) this may result in absurdly low rarefaction sums for some samples. Absurd, because each pollen-diversity estimate has its rarefaction sum as the upper limit. The lowest rarefaction sum for 1 mm² occurring in our data set is 10, and the much lower pollen-accumulation values per mm² per yr imaginable in some other regions will require a larger accumulation surface for estimating accumulation-based pollen diversity. Nevertheless, appropriate regional standards in round numbers (eg, 1, 10, 100 mm²) would facilitate comparisons among sites.

Pollen diversity estimated on pollen accumulation is unaffected by evenness, so it is more a measure of taxonomic richness in the landscape. It is not a perfect measure for many reasons, most of which it shares with constant-count pollen diversity, eg, pollen types are not one-to-one equivalent to plant taxa, plants may not flower for climatic or other reasons, the chance of finding pollen produced by low pollen producers is anyway smaller than that of large pollen producers, and the effectiveness of pollen transport to the studied sediment may change when climate changes (wind, precipitation), the size of the studied lake or mire changes, or the structure of vegetation changes (eg, appearance of forest in a formerly unforested landscape, filtering away pollen from low-growing plants). Pollen diversity estimated on pollen accumulation also has specific problems not shared by constant-count pollen diversity, namely those related to the accumulation of the sediment, such as sediment focusing (causing a spatially uneven accumulation of the pollen assemblages) or to uncertainties in the depth–age model. Nevertheless, accumulation-based pollen diversity can be interpreted in terms of taxonomic richness in the landscape if due caution is taken.

In conclusion, accumulation-based estimates of pollen diversity can be interpreted in terms of species richness in the landscape, whereas trends in constant-count pollen diversity might be explained but cannot be interpreted.

A possible future development: species turnover and pollen-accumulation rates

Species turnover in the landscape depends on appearance and disappearance of plant species, and is independent of the abundance of individual species. For estimating species turnover in the landscape from pollen assemblages, pollen percentages are therefore unsuitable because of the evenness problem discussed above. Instead, pollen-accumulation rates may be used effectively for estimating species turnover in the landscape with the following steps:

- Calculate for each pollen sample the surface represented by the total pollen count of interest.
- Select the smallest surface among the pollen samples.
- Calculate or estimate for each pollen sample which pollen types were deposited on this smallest surface.
- Use these presence/absence data to estimate turnover of pollen types, according to an algorithm of choice. Software for this work is not readily available, but might be developed by a skilled person.
- The final step, from turnover of pollen types to turnover of taxa in the landscape, is not trivial and will need much care and thought (Odgaard, 2001).

The potential sources of error are the same as in the accumulation-based pollen diversity.

Other summarizing techniques in palynology affected by evenness

All techniques of summarizing pollen assemblages that use percentages (as opposed to presence/absence) are sensitive to evenness. If the results are to be interpreted in terms of vegetation, it is relevant to note that differences in pollen productivity among taxa modify the evenness in the pollen assemblages compared with the evenness in the vegetation that produced the pollen. The consequences of this are considered here for a few summarizing techniques.

Rates-of-change in a sequence of pollen assemblages are commonly estimated with RATEPOL (J.M. Line and H.J.B. Birks, unpublished software, 1994) and are based on pollen percentages. If the aim is to interpret the results in terms of rates of change in the vegetation, the problems may seem the same as in constant-count pollen-diversity estimates: evenness of the pollen assemblages dominates the result, and among-taxa differences in pollen productivity modify the evenness in pollen assemblages compared with the evenness in vegetation. But contrary to estimates of diversity, rates of change estimates are intended to reflect evenness. However, the evenness should be that of the vegetation, not that of the pollen assemblage if that is different. Therefore, although estimating the rates of change on the basis of pollen-accumulation rates (grains/cm² per yr) might be possible, this solves the problem only in part: percentage effects are removed, but the among-taxa differences in pollen productivity will continue to affect evenness and therefore the estimated rates of change. The solution, the only one I currently see, is to divide all counts of individual pollen types by their estimated pollen productivity, prior to estimating rates of change. This requires pollen-productivity estimates for all pollen types in the assemblage. This is a very distant goal, and until then the link between estimated rates of change in the pollen assemblage and rates of change in the vegetation is undefined and unstable.

Also numerical zonations of pollen assemblages (Birks and Gordon, 1955; Bennett, 1996) use pollen percentages, so the results are dominated by evenness. If the zonations are to be interpreted in terms of shifts in the vegetation, the problems are the same as in rates of change estimates. The solution might be the same, ie, application of pollen-productivity estimates.

Ordination techniques such as principal-components and redundancy analyses use abundances of pollen types and might therefore be sensitive to the differences in pollen productivity among plant taxa. We should therefore be careful with the interpretation of such analyses in terms of past vegetation.

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